

The Impact of PTAs' Environmental Provisions on Chinese Firm-Level Export Performance

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Research Question & Data

Goal: does the environmental content of China's PTAs shift exports towards green goods and away from dirty goods?

- ▶ Chinese customs transactions, 2000–2015 (45.8M obs.)
- ▶ 14 PTAs signed by China (23 treated destinations)
- ▶ EP content: World Bank DTA & TREND databases
- ▶ Green/dirty goods: OECD CLEG list & pollution-intensity mapping
- ▶ Robustness samples: HS4-restricted comparison, CEM-matched destinations

Empirical Analysis

$$\ln x_{fpdt} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + FE + \varepsilon_{fpdt}$$

firm $\times$ product $\times$ destination $\times$ year, with firm–destination–year
fixed effects

Three levels of inference

- ▶ Asymptotic
- ▶ Wild cluster bootstrap
- ▶ Permutation

Robustness of measurement

- ▶ WB *and* TREND indices
- ▶ Continuous CO₂ intensity

Results: The Triple-Difference Estimates

	Full panel		Collapsed panel	
	WB	TREND	WB	TREND
$EP \times \text{green}$	-0.0023	-0.0001	-0.0046	0.0018
Bootstrap p	0.69	0.98	0.65	0.39
$EP \times \text{dirty}$	-0.0044*	-0.0009	-0.0119***	0.0004
Bootstrap p	0.19	0.18	0.07	0.86
Permutation p	—	—	0.28	0.82

- **Green:** a robust null
- **Dirty:** negative, but fragile to inference method

Results: The Extensive Margin (PPML)

Zero-filled HS6–destination–year grid — do agreements create *new* trade flows?

	WB index	TREND index
$EP \times$ green	0.0015	0.0001
Asymptotic p	0.74	0.95
$EP \times$ dirty	-0.0301*	0.0028
Asymptotic p	0.06	0.55

- ▶ No new green trade — same null as the intensive margin
- ▶ Dirty margin: same pattern, marginal under one index only

Takeaways

- ▶ **Green margin: a robust null**
- ▶ **Dirty margin: fragile**
- ▶ **Content over presence**

Next steps

Preferential tariffs · Webb-weight inference · submission
by year-end

Thank you